



How Playwright's Architecture Actually Works



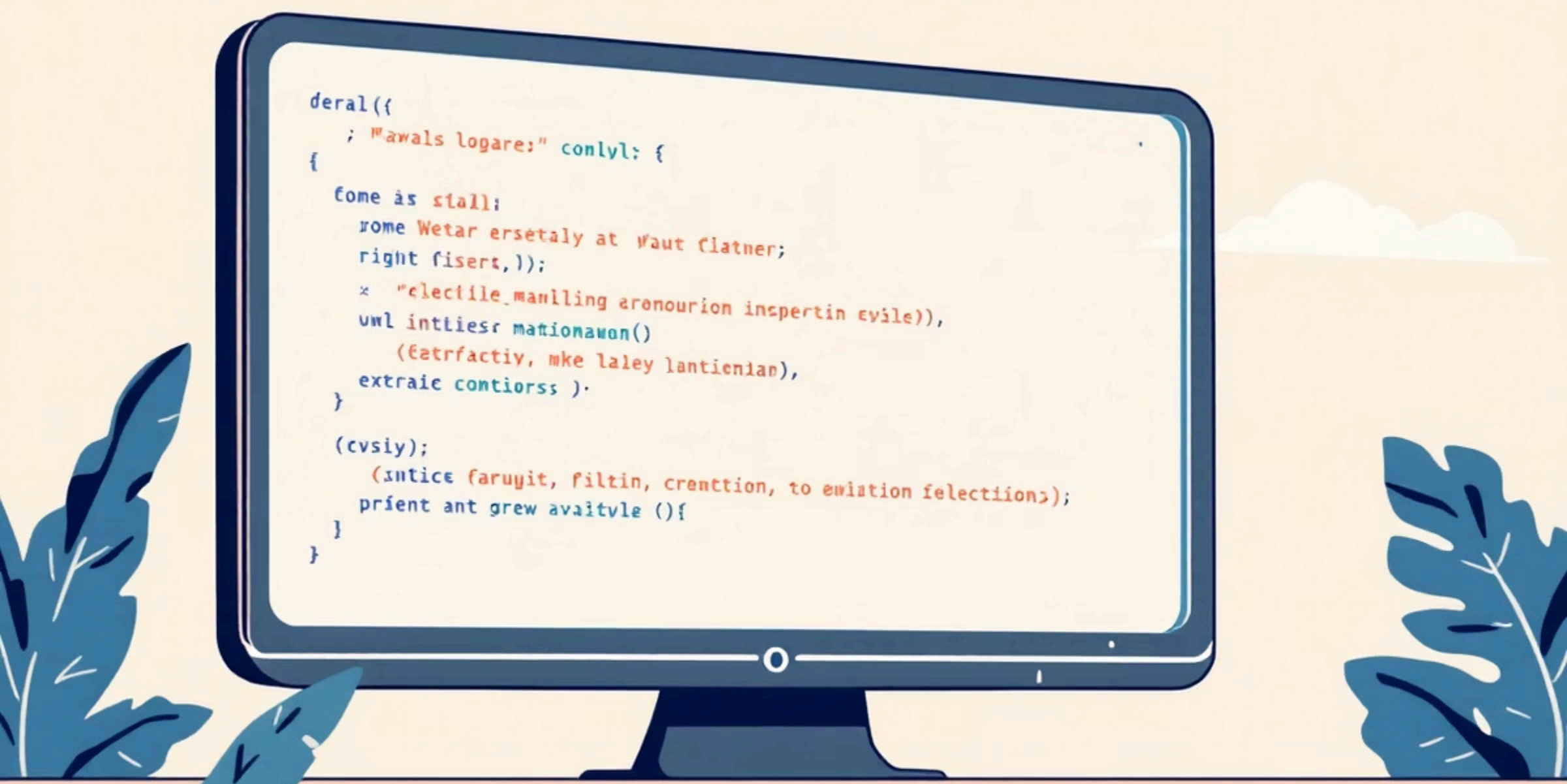
Playwright

Understanding the technical foundation behind
Playwright's reliable browser automation capabilities



The Big Picture

Playwright uses a **persistent WebSocket connection** between your test code and browser engines. This architectural choice eliminates the connection overhead that plagues traditional tools like Selenium.



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Client Layer: Your Test Code

Multi-Language Support

JavaScript, TypeScript, Python, Java, .NET

Test Commands

Element interactions, page navigation, assertions



The Game Changer: WebSocket Connection

Single, persistent WebSocket connection
established at test run start - no reconnection
overhead like traditional HTTP-based tools

Playwright Server: The Command Center

01

Receives Commands

Client sends test instructions via WebSocket

02

Translates Instructions

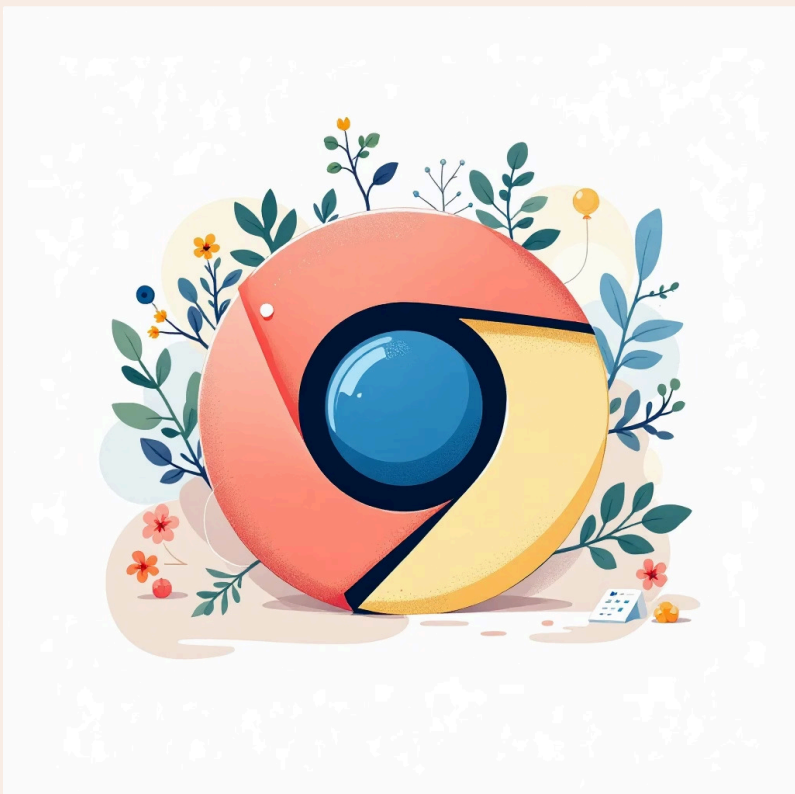
Converts to browser-specific protocols

03

Executes Actions

Communicates directly with browser engines

Browser Communication Protocols



Chromium

Uses Chrome DevTools
Protocol (CDP) for
direct browser control

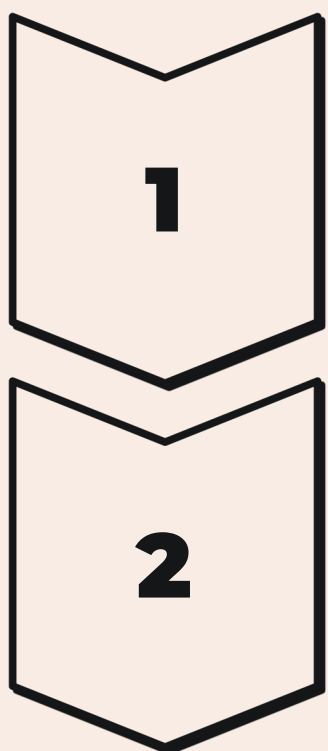


Firefox & WebKit

Custom protocols
implemented by
Playwright team, similar
to CDP functionality



Bi-Directional Communication Flow



Client → Server

Test commands sent over WebSocket

Server → Client

Real-time responses and events streamed back

Why Persistent Connections Matter

0

Connection Overhead

No repeated handshakes during test execution

↓85%

Latency Reduction

Compared to HTTP-based automation tools

99.9%

Stability Improvement

Eliminates connection-related flakiness



Test Execution Lifecycle

1

Connection Established

WebSocket opens at test run start

2

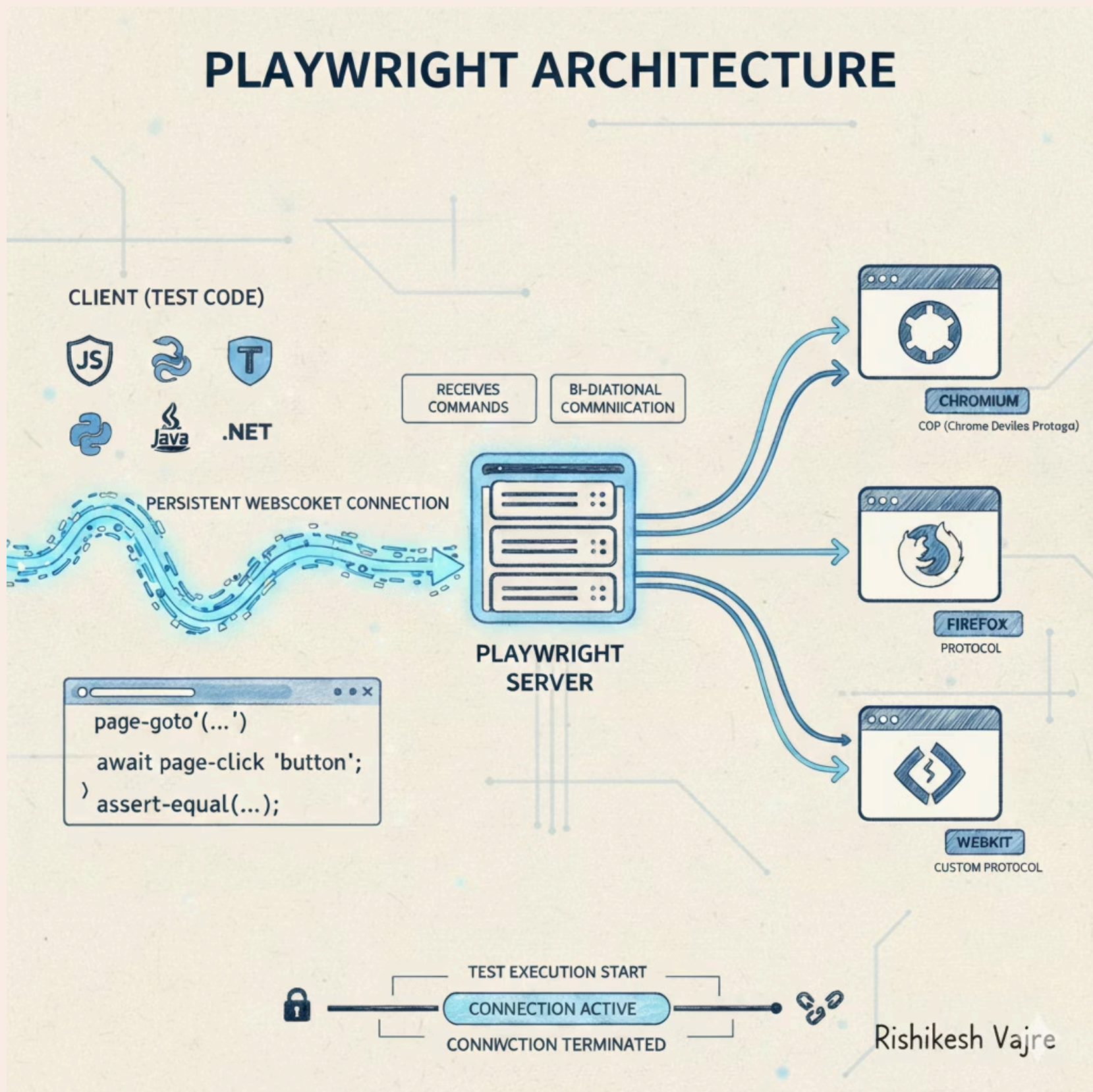
Commands Executed

Sequential test operations over persistent connection

3

Connection Terminated

Clean disconnection after test completion



Architecture Advantage Unlocked

Playwright's WebSocket-based architecture delivers the reliability and speed that modern test automation demands.

Share this breakdown with your QA team!